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Department of Electronics and Communication Engineering

EC8811- PROJECT WORK

Project Title

**SMART AUTONOMOUS VACUUM CLEANER ROBOT
FOR COTTON INDUSTRY**

Project Guide

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Abstract:

The rise of automation in industrial environments has prompted the development of innovative solutions to enhance operational efficiency and cleanliness. This project covers the design and implementation of a **SMART ROBOT for industrial cotton floor cleaning**, specifically adapted for textile production facilities. The robot **integrates sensor technology, machine learning algorithms, and real-time data processing** to autonomously navigate and clean the floor areas, effectively addressing the challenges posed by cotton waste.

We **used image processing to find the cotton** on the floor so that the smart robot vacuum and collect it up. This SMART ROBOT not only enhances operational efficiency but also contributes to a sustainable industrial ecosystem **by minimizing manual labour and optimizing resource usage**.

Base Paper

- 1. Developing an Unmanned Rover for Garbage Management System using Internet Protocol of Raspberry PI** - concept we adapt from this paper is project highlights technology's essential role in improving waste management.**(we do cotton collection management)**.
- 2. Smart Vacuum Cleaner** - concept we adapt from this paper is An automatic vacuum cleaner robot is an electronic device that is programmed to clean the floor using a vacuum cleaning mechanism. **(Sensors concept)**
- 3. Design and Analysis Of Automatic Robotic Vacuum Cleaner** - concept we adapt from this paper code is used to show the simulation of robot moving in a user-defined path. The robot navigates front with a castor wheel and two rear wheels. **(Robot movement)**

Introduction

Design and implement an automated vacuum cleaner robot that efficiently performs floor cleaning tasks **under the heavy machineries with minimal human intervention**. The robot must be capable of navigating various indoor environments and managing its operational resources autonomously.

The increasing demand for efficiency and cleanliness in industrial settings has led to a surge in research and development of autonomous cleaning systems. Specifically, the **textile industry**, with its unique challenges related to lint management, necessitates innovative solutions.

Objective

To develop and deploy an automated vacuum cleaner robot in cotton mills, designed to efficiently **clean and remove waste cotton** from the floor, thereby enhancing workplace safety, reducing manual labor, and maintaining a clean operational environment.

Observation

Cotton mills are integral to the textile industry, where raw cotton is processed into yarns and fabrics. During this process, various forms of waste are generated, including **leftover cotton** from production processes. Understanding and **managing waste cotton** is crucial for improving operational efficiency, reducing environmental impact, and exploring potential for resource recovery.

Motivation

- To reduce the Environmental Impact like Fire accident (caused by thermal heat energy), reduce Pollution.
- To give effective and Economic Benefits like recycling processes.
- To improve the product efficiency.
- To reduce the frequency with which workers are required to clean four times during a day shift and three times during a night shift, consider implementing changes.
- To improve the TQM(Total Quality Management).

Software

- Python
- Raspberry pi OS

Proposed Method

We have integrated the normal Vacuum Cleaner Machine with **sensors** (used to move the robot by identify the obstacles), **camera** (used to identify the cotton by image processing).To collect the waste cotton around the textile industry.



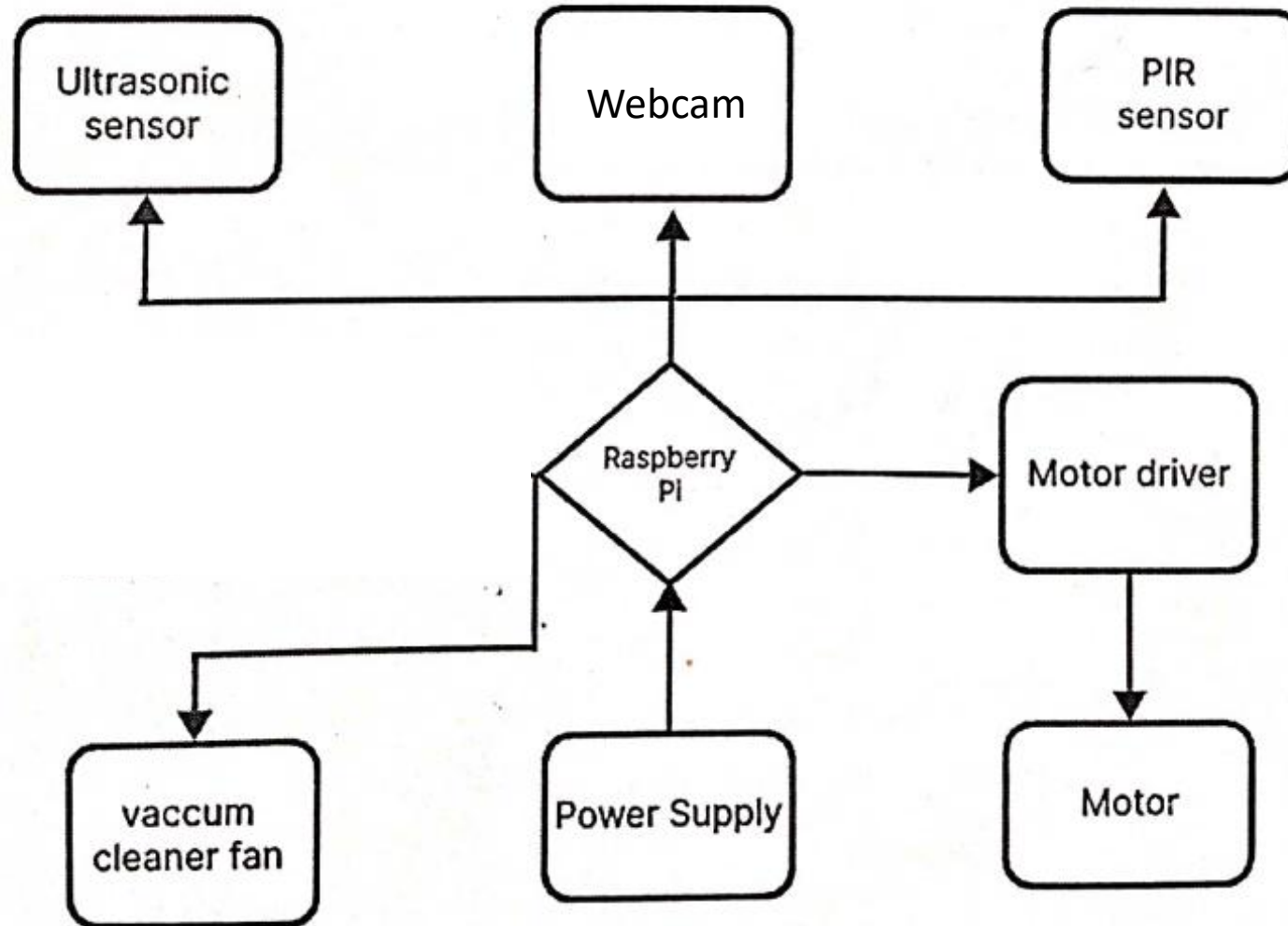
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**Image
Processing**

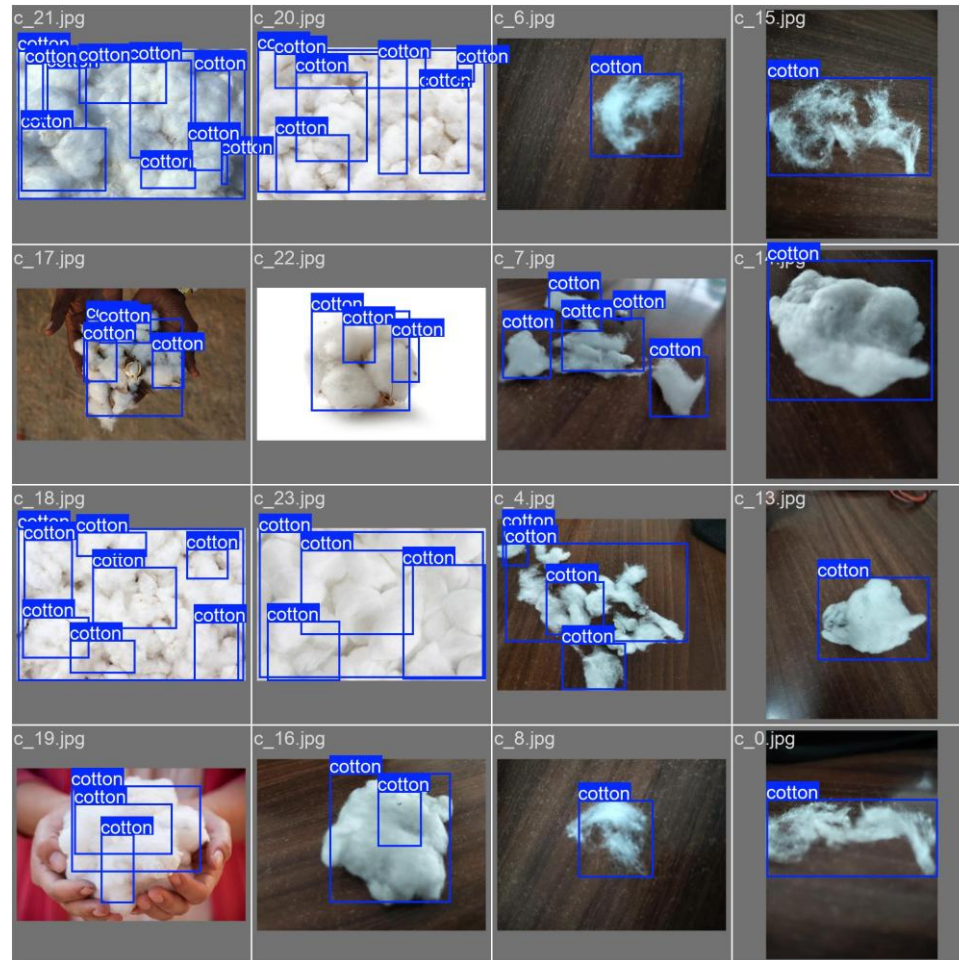
Block Diagram of the Proposed System



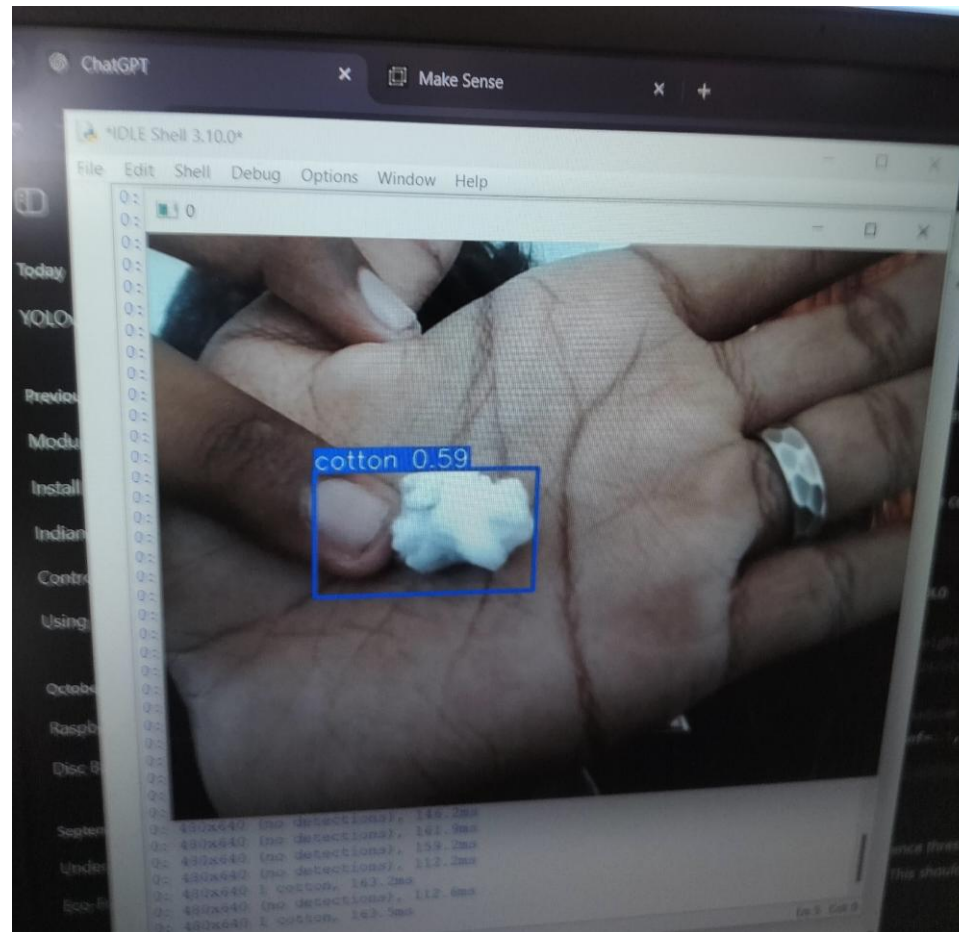
Working of Proposed Method:

- We have trained the image processing dataset of cotton in yolov8 model in raspberry pi 5 & the webcam used to detect the cotton.
- The robot fully function the motor driver to move the robot to the cotton.
- The ultrasonic sensor detect any abstracle for the robot to move forward.
- The PIR sensor used to detect the human interference & the motor stops.

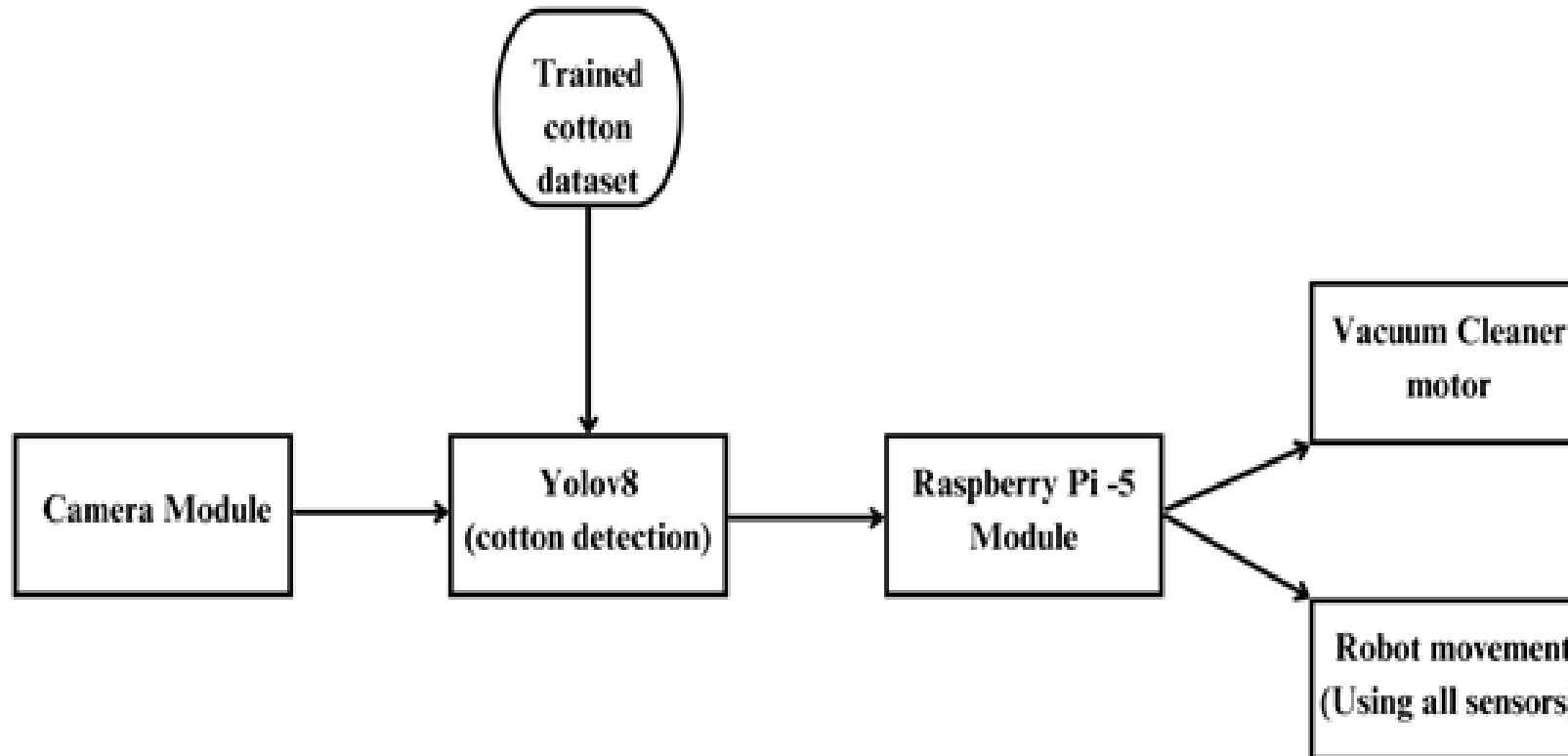
Dataset Batch Models



Cotton identified



Block Diagram of the Image processing



Prototype Structure



Result



https://colab.research.google.com/drive/1XysbZM3lpyurwCXzsXRMjrBw84g7t6sn?usp=drive_link

https://drive.google.com/drive/folders/1Qxa_Zz_ddytACiGht4lxY59nOaWUayMj?usp=drive_link

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THANK YOU